Discussion note. Not peer reviewed.

State the result in the title

A discussion note on exhibit sentences

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Abstract An exhibit title that names the topic leaves the reader to invent the sentence. A title that states the comparison can be checked against the picture. This note argues for the second habit and does not report an experiment.

Keywords exhibits; graphical perception; titles

1 The claim

A chart titled “Regional costs” tells the reader where to look and not what was found. A chart titled “Overlap zones cost more per stop than hub zones” makes a claim the picture can support or embarrass. The second title is harder to write, which is the reason to prefer it. The title is part of the evidence, not a label beside it. This is a writing claim. It is not a finding about memory, and it is not attributed to a laboratory. The only empirical work cited below is about how people decode graphics, not about how they remember sentences.

2 What the sentence is doing

A topic title and a claim title do different jobs. The first names the slice of the world. The second names the comparison and, usually, its direction. Once the direction is in the title, a reader can ask a narrow question: does the picture show that direction, in that unit, for those categories?

Title	What the reader must supply
Regional costs	Which regions, which direction, which unit
Overlap zones cost more per stop	Whether the bars agree

Table 1: Two titles for the same picture.

The second row is still not a proof. It is a sentence with a truth value. That is the property worth buying. A title with no truth value cannot be wrong, and a title that cannot be wrong is not doing analytical work.

3 What to encode

Cleveland and McGill ordered elementary perceptual tasks by how accurately people extract quantitative information. Position along a common scale sits at the top of that order; area, volume, and color saturation sit lower (Cleveland and McGill, 1984). For a comparison of a few categories, a bar or a dot on a shared axis is the encoding that matches the task. A pie asks the reader to compare angles or areas for a job that was a comparison of lengths. Tufte’s discussion of graphical integrity and of data-ink is the other standing reference for this habit: the picture should show the data, and the labels should say what the data are, in the unit the data were measured in (Tufte, 2001). Neither citation is a license to put an invented number under a careful title. A precise title on an illustrative series still has to say that the series is illustrative.

4 A worked exhibit

Suppose four fulfillment zones and a cost per stop. The numbers below are a sketch, chosen so the comparison is obvious, and they are not measurements.

Zone	Yuan per stop	Read as
Hub-served	6.4	Baseline
Single coverage	7.1	Close to the hub
Overlap	9.8	The claim
Remote	11.2	A different problem

Table 2: Illustrative cost per stop, yuan. Not a measurement.

The honest title is “Overlap costs more per stop than hub-served zones in this sketch.” The title “Cost by zone” would also fit a page on which the remote zone was cheapest. That is the test: if the title would survive a reversal of the data, it is not yet a result.

5 What this note does not show

There is no sample of readers, no response time, and no test of memory. Cleveland and McGill studied decoding of quantities, not recall of titles. Nothing here measures whether a claim title changes what a person remembers a week later. A study of that question would need a design this note does not have. The note also does not say that every heading in a paper should be a finding. A methods section can be titled “Methods.” The pressure applies where a picture is being asked to carry an argument: the title of that picture, and the title of the slide or the subsection that exists only to present it.

1. No new dataset was collected.
2. No DOI is attached, because none was re-checked for this note.
3. The yuan figures in the table are illustrative and labeled as such.

References

Cleveland, W. S., and McGill, R. (1984). Graphical perception: Theory, experimentation, and application to the development of graphical methods. Journal of the American Statistical Association, 79(387), 531–554.

Tufte, E. R. (2001). The Visual Display of Quantitative Information (2nd ed.). Cheshire, CT: Graphics Press.